

Intelligent Food Waste Prediction and Redistribution System Using Machine Learning

Harikumar M.

*Department of Computer Science
and Engineering*
Rajalakshmi Engineering College
Chennai, India
harikumar.m@rajalakshmi.edu.in

Sowndariya R.

*Department of Computer Science
and Engineering*
Rajalakshmi Engineering College
Chennai, India
230701329@rajalakshmi.edu.in

Smith N. S.

*Department of Computer Science
and Engineering*
Rajalakshmi Engineering College
Chennai, India
230701323@rajalakshmi.edu.in

Elumalai B.

*Department of Computer Science
and Engineering*
Rajalakshmi Engineering College
Chennai, India
230701084@rajalakshmi.edu.in

Mathankumar M.

*Department of Computer Science
and Engineering*
Rajalakshmi Engineering College
Chennai, India
230701181@rajalakshmi.edu.in

Abstract—Food waste in college cafeterias can be attributed to student attendance uncertainty, inaccurate meal demand estimation, and the lack of organized surplus redistribution. Traditional methods of food preparation rely either on experience or rough estimates of student attendance, leading to either surplus or insufficient food preparation. This article describes an innovative system of predicting and redistributing food waste in college cafeterias. The system incorporates student voting, machine-learned demand estimation, and surplus monitoring and NGO coordination mechanisms to help reduce surplus food. Students can vote on a web portal for their breakfast and lunch needs. These results, past consumption data, customer numbers, food preparation, seasonal adjustments, holidays, and events can be used to predict demand. The estimated number is compared to the number of meals planned to be prepared to estimate the amount of surplus food. After meals, cafeteria workers can monitor the amount of surplus food recorded in the system. Finally, the application can notify registered nearby NGOs in case of edible surplus to collect and redistribute food to those in need. The described system had food demand estimation, voting, surplus tracking, NGO identification, and collection monitoring subsystems. While the paper demonstrated the predicted number of meals and voting results, there are no numerical results to see how well the model performed or reduced food waste in practice. More research will be required to evaluate the system's practical performance with real cafeteria demand data. The paper's contribution is that it provides the framework for tying food demand estimation to a surplus redistribution system and NGOs.

Index Terms—Food waste management, machine learning, demand forecasting, cafeteria management, food redistribution, surplus food, sustainability, NGO coordination.

I. INTRODUCTION

Food wastage in educational institutions emerges as a significant issue due to the lack of synchronization between the supplied quantity of food and the consumption demand

of the students. College and university messes suffer from the inaccuracy of meal planning as they prepare the food keeping in mind only approximated figures of attendance. It creates problems including wastage of food, utilization of extra resources, and additional expenses for the messes along with adverse environmental effects. Hence, there is a need for an efficient and effective system for the better estimation of meal demand and management of food resource planning. This paper proposes the FoodPulseAI project, a smart food-demand management and student meal-voting system, for educational institutions. The system allows students to provide their voting preferences about the meals through their web interface. Students would provide their voting input for their respective breakfast and lunch timings; thus, helping the messes in having an exact estimation of the expected number of consumers. The website will keep track of all the students' information and voting preferences in a database. It will also provide a dashboard for the messes for tracking the voting data of students and managing all food-related activities of the institution. The voting data of students can be helpful in determining the meal patterns for the mess. Through the collective participation of all students, the project will ensure decreased wastage of food. The system will keep track of managing meal schedules, surplus foods, and other relevant activities related to food. Thus, through smart technologies, the paper proposes a food management system for educational institutions. Keeping in mind the benefits of estimating the needs of the students, the project will help the messes in transforming their traditional planning system towards a modern and smart one. Consequently, the primary focus of the development of this project is to provide a digital platform between the students and messes for effective food management. It will help in ensuring a sustainable use of resources at college and

university levels and provide a better mechanism for managing food resource planning.

II. LITERATURE SURVEY

Food waste has become an essential issue in institutional and large-scale food-service settings, where miscalculations of meal demands can lead to excessive amounts of food being prepared and wasted, increasing resource wastage and operational expenses. Recent studies have investigated the use of machine learning, digital platforms, demand forecasting, food-rescue systems, and intelligent redistribution to deal with these difficulties. Shi et al. [1] introduced a bandit data-driven optimization strategy for crowdsourcing food rescue platforms, demonstrating how data-driven optimization can enhance food-rescue operations and volunteer engagement. Their work stresses the value of using continually generated platform data to increase the effectiveness of food recovery operations. In a similar vein, Wan et al. [2] proposed a machine-learning framework for compost maturity prediction. Using feature analysis and prediction modeling, their research demonstrated how machine learning can be utilized to automate food-waste processing and assist intelligent waste-management systems. Suresh et al. [3] introduced the ZeroHunger digital solution that employs a combination of AI/ML, web application, Firebase, food quality assessment, and donor-recipient coordination to coordinate food donations with organizations in real time. Kumar et al. [4] investigated food-waste prediction using Random Forest, geolocation, and automated notifications to connect food donors to nearby NGOs. These studies have demonstrated that intelligent prediction and digital coordination can aid both food-waste reduction and redistribution. Kim and Yoon [5] investigated the use of machine-learning techniques to predict food-waste quantities in military dining facilities. Their work used environmental factors and meal composition to indicate that machine-learning models can be applied in large-scale food-service settings to estimate waste quantities. Faezrad et al. [6] investigated university dining systems and introduced an Artificial Neural Network (ANN)-aided model using meal reservation data and student show/no-show behavior to predict demand and reduce surplus food. Eriksson et al. [7] investigated machine-learning-based guest attendance prediction in the context of Swedish public catering. Their work indicated that predicting expected attendance can assist in more appropriate food preparation and thus reduce food waste. Rodrigues et al. [8], investigated the short-term food-demand forecasting using historical catering data, indicating the potential that machine-learning-based demand forecasting has for the estimation of meal requirements and waste-reduction strategies. Jung and Kim [9] introduced machine-learning predictive models and a web application for university cafeteria attendance, using XGBoost and Random Forest models to predict breakfast, lunch, and dinner attendance. This work is particularly relevant to educational institutions as it demonstrates how cafeteria attendance prediction can be enhanced using a web-based system. Turker [10] introduced a data-driven approach to campus dining, emphasizing the use of

demand prediction and sustainability. The work highlighted the potential that campus-related data analytics have for enhancing food-planning decisions and reducing food waste.

In addition to demand prediction, computer-vision approaches have been investigated for the detection of food waste. Li et al. [11] introduced a machine-vision based approach for the automatic identification of food waste in university cafeterias using deep learning and plate-image recognition. This demonstrates the applicability of artificial intelligence not only in the field of food-demand prediction but in the measurement of actual food-waste as well. Agmapisarn et al. [12] investigated surplus-food redistribution through digital platforms, emphasizing real-time food listings and collaboration with charitable organizations. Their work demonstrates the key role of accessibility and real-time information in effective surplus-food redistribution. Shivani S. S. and Usha M. [13] introduced a web-based food donation and redistribution system using a centralized database, location tracking, and automated notifications. The system aims to enhance the coordination between food donors and NGOs, facilitating timely collection of surplus food. Devika et al. [14] expanded on this concept and introduced an IoT-enabled surplus food rescue platform using sensors, Firebase, web technologies, and NGO notifications. These systems demonstrate the importance of the integration of monitoring and communication technologies to food-rescue operations. Agmapisarn et al. [15] emphasize on the role of mobile and digital platforms in facilitating surplus-food sharing. Their work demonstrates the potential that real-time food listings and sustainability-oriented features have for encouraging food redistribution and food-waste reduction. Hübner et al. [16] investigated the utilization of machine-learning-based demand forecasting in combination with life-cycle environmental assessment in a bakery case study. Their findings demonstrate that enhanced demand forecasting can reduce unnecessary food returns while benefiting the environment. Food redistribution can be viewed as an optimization problem as well. The integrated foodbank network design discussed in [17] employs Mixed-Integer Programming to coordinate donors, foodbanks, recovery agencies, and beneficiaries. The case study presented here indicates the potential that network-level optimization has on the enhancement of the efficiency and equity of food redistribution services. Oroski and da Silva [18] through a review of food-waste-reducing platforms, have emphasized the rising role of digital food-sharing platforms in connecting surplus-food suppliers with recipients, monitoring food waste, and generating useful information for the purpose of food recovery. Murthy et al. [19], introduced a web-based surplus-food redistribution system that records surplus-food information and generate location-based alerts for NGOs, volunteers, and collection organizations. This demonstrates the usefulness of online coordination in the enhancement of the speed and efficiency of surplus-food collection. Turker [20], further emphasized on the utilization of data-driven campus dining, employing machine-learning regression and campus attendance/dining data in the prediction of daily meal demand.

III. PROPOSED SYSTEM

The proposed FoodPulseAI is an intelligent web-based food-demand management system, aimed at reducing the food wastage in campus cafeterias by predicting the daily meal requirement. The system exploits the real-time student meal voting, historical cafeteria data, contextual information, machine-learning based demand prediction, administrative monitoring and surplus-food redistribution in a combined framework. Unlike the ordinary cafeteria planning methods that base on the estimated attendance or historical consumption, FoodPulseAI captures the students' meal requirements before food preparation. Students can independently indicate their requirement of breakfast and/or lunch, while the administrators can monitor the collected responses and use the resulting demand information for food-planning decisions.

The proposed system comprises the student-side interface, administrative interface, backend processing layer, centralized database, demand prediction module, and surplus-food management module.

A. Student Module

The student module offers registration, authentication, and meal-voting functionality. A new student can register using the required student details and then access the system using valid credentials. After a successful login, the student is redirected to the meal-voting interface. The student is not expected to select individual food items. Instead, the system addresses the requirement of each meal.

The voting interface offers:

Breakfast - Yes/No
Lunch - Yes/No

Breakfast and lunch are considered as independent voting activities. Therefore, a student can submit a breakfast requirement without submitting a lunch response, or vice versa. This simplifies the participation of students while generating meal-specific demand information.

B. Real-Time Meal Demand Collection

Each student response is associated with the student's identity, meal type, voting date, and submission information. The response is validated by the backend before being stored in the centralized database.

The number of students requesting meal m is calculated as:

$$V_m = \sum_{i=1}^N I(v_{i,m} = \text{Yes})$$

where V_m represents the current demand signal for meal m , N represents the participating students, and $I(\cdot)$ is an indicator function.

C. Administrator Module

The administrator module offers centralized control over the food-management process. The administrator can access the dashboard to monitor student participation and meal requirements. The dashboard presents the current breakfast and lunch voting information, allowing administrators to monitor changes in expected demand.

D. Voting-Time Management

The administrator can define the permitted voting period. The backend verifies the configured date and time before accepting a student response. This prevents voting outside the permitted period and provides administrators with control over the data-collection process.

E. Machine-Learning Demand Prediction

The proposed system combines the real-time student voting with historical cafeteria data and contextual factors for demand prediction.

Historical data can include:

Prepared quantity
Consumed/sold quantity
Surplus quantity
Meal type
Date

$$X = [V_m, D, M, T, R, H, E] \quad (1)$$

where V_m is the current student vote count, D is the day of the week, M is the month, T is the temperature, R is the rainfall, H is the holiday indicator, E is the examination indicator.

$$\hat{Y} = f(X) \quad (2)$$

where f represents the trained prediction model and $\hat{Y}$ represents the estimated meal demand.

F. Food Preparation Recommendation

The predicted meal requirement is converted into a recommended preparation quantity. To compensate for prediction uncertainty and reduce the potential for insufficient food, the proposed system applies a 10 percent safety factor.

$$Q_{\text{prep}} = \hat{Y} \times 1.10 \quad (3)$$

For example, if the predicted demand is 100 meals, the recommended preparation quantity is:

$$Q_{\text{prep}} = 100 \times 1.10 = 110 \quad (4)$$

The safety factor can be adjusted based on institutional requirements and observed prediction performance.

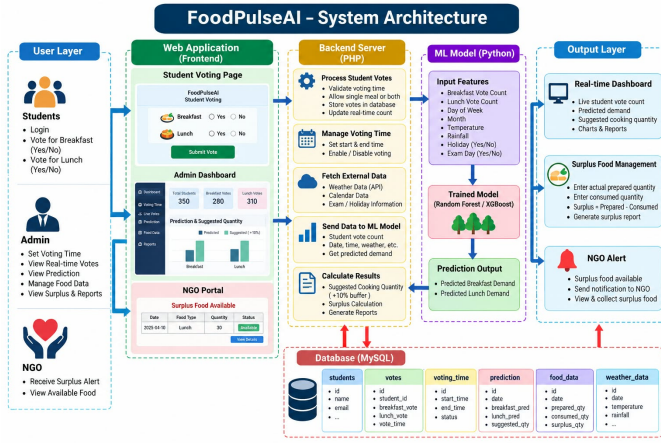


Fig. 1. System Architecture

G. Surplus Food Management

After food preparation and distribution, actual prepared and consumed quantities can be recorded. The surplus quantity is calculated as:

$$Q_{\text{surplus}} = \max(0, Q_{\text{prepared}} - Q_{\text{consumed}}) \quad (5)$$

If the resulting surplus is greater than zero, the system identifies the meal as surplus food. This creates the feedback mechanism between demand prediction and actual cafeteria outcomes. The recorded consumption and surplus information can then become part of the historical dataset for future demand prediction.

H. NGO and Redistribution Module

The NGO module keeps information about organizations that can support surplus-food redistribution. When suitable surplus food is identified, administrators can coordinate with registered NGOs.

Surplus Detection → NGO Identification → Notification
→ Collection → Redistribution (6)

This ensures that the proposed system does not focus only on preventing excess food preparation but also offers the mechanism for managing unavoidable surplus food.

IV. METHODOLOGY

The FoodPulseAI methodology is characterized by a series of steps of collecting, preprocessing, predicting, evaluating, and using meal-demand data. It will be based on student votes in real-time and historical cafeteria data.

A. Data Collection

The first step is to collect data. In this example, it will be done in real-time based on student votes. The authenticated students will be able to vote if they need breakfast and/or lunch. Their votes will be stored in a database along with the student ID, meal type, and date. On the other hand,

the historical data will contain the cafeteria data, such as food prepared, consumed, and surplus, meal type, and the date. Other contextual variables are the day, month, holiday, examination, temperature, and rainfall.

B. Data Preprocessing and Feature Engineering

The second step is to preprocess and engineer the data. The database will be cleaned and formatted in the machine-readable form. The variables will be converted into features, namely the day, month, holiday, examination, temperature, rainfall, the number of students who voted for a particular meal as the input feature, and the total food demand or consumption as the target variable.

C. Machine-Learning Model Development

The study will consider two (2) machine learning models, namely, Random Forest Regression and XGBoost Regression to predict demand for meals. The historical data will be used to train the model. It will use the input features and the target variable identified

D. Model Evaluation

The next step is to evaluate the model. The study will generate the Mean Absolute Error (MAE) given by the equation below:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (7)$$

where Y is the actual demand and $\hat{Y}$ is the predicted demand. The model with the lowest error will be used for the final set of data.

E. Demand-based Food Preparation

Using the predicted demand, the food preparation quantity will be calculated as: followed by equation (3)

This will give a safe percentage of 10 for food preparation with some leeway for errors.

F. Surplus Detection and Redistribution

After food distribution, the study will calculate the surplus using the following equation (4). The information will be stored in the system and shared with the NGO coordination module. The surplus detection process will involve steps 1 through 5 above. The following workflow will be followed:

Surplus Detection → NGO Selection → Notification → Collection → Redistribution

The actual consumption and surplus may also be stored as part of the historical data set for future demand analysis.

G. Real-Time Demand Updating

The process will be repeated daily during active voting to update the demand based on the latest student preferences. This can be represented by the following equation:

Student votes → Database → Feature Generation → ML Model → Demand Prediction → Preparation Recommendation

The above workflow will ensure that FoodPulseAI uses real-time student demand and historical demand patterns to inform food planning and preparation.

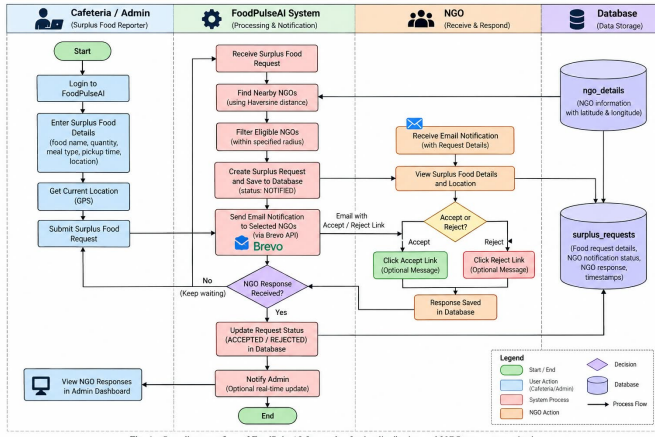


Fig. 1. Overall system flow of FoodPulseAI for surplus food redistribution and NGO response monitoring.

Fig. 2. Flow Chart

V. RESULTS AND DISCUSSION

The FoodPulseAI system allows for a complete collection of students' meal requirements, estimating the demand for food at the cafeteria, and managing surplus. The results of these processes are discussed in terms of student voting, demand prediction, food preparation, and surplus management.

A. Student Voting Results

The student voting function allows students to choose their preferences regarding the breakfast and lunch individually. For the voting process, students are required to log in to the system using the authentication module in order to make their choice. The results of the voting are collected and stored in a common database, separated into the data for breakfast and lunch.

Meal Type	Voting Options	Result
Breakfast	Yes / No	Breakfast demand count
Lunch	Yes / No	Lunch demand count

Table 1: Meal Demand Voting

The number of votes "Yes" can indicate the demand for each particular meal. Therefore, using this method, administrators can control the students' needs without estimating them manually.

B. Demand prediction results

The information about votes can be used along with the past data of the cafeteria demand and specific features to predict future demand. For this purpose, Random Forest and XGBoost Regression algorithms can be implemented.

Model	Input Features	Target	Evaluation
Random Forest	Votes, historical and contextual data	Meal demand	MAE
XGBoost	Votes, historical and contextual data	Meal demand	MAE

Table 2: Demand Prediction

The Mean Absolute Error (MAE) is used to compare the prediction models. A lower MAE indicates that the predicted demand is closer to the actual consumption. Therefore, the model with the lower MAE can be selected for demand estimation.

C. Food Preparation Results

The anticipated demand is modified into a recommended quantity of preparation using a 10percent safety factor. This will give the cafeteria staff enough but limited food for preparation with some leeway for the expected demand.

D. Surplus Food Management

The surplus is then generated by comparing the served quantity with the prepared quantity. This surplus is recorded and reported as the amount of food wasted while preparing meals and is reportable to the NGO coordination module.

Stage	Information
Food Prepared	Total quantity prepared
Food Consumed	Quantity consumed
Surplus Detected	Remaining quantity
NGO Selection	Suitable registered NGO
Notification	Surplus availability communicated
Redistribution	Surplus transferred for further use

Table 3: Food Management

E. Discussion

The analysis shows that the FoodPulseAI connects the current student turnout, which feeds back into the cafeteria demand forecasting and management module, and the surplus management. On the one hand, the student voting provides up-to-date information on the demand, while the historical and contextual data provide the basis for the prediction. On the other hand, the surplus module also offers an additional benefit of a surplus management system.

VI. CONCLUSION AND FUTURE SCOPE

A Food-Waste Prediction and Management Model For School Canteens A Solution To The Needs Of Student Using Analytics Essay A solution to the needs of students should be developed by considering ways of addressing the requirements and reducing wastage in institutes. Using the proposed solution, students will virtually provide their demands for breakfast and lunch. The two responses will help the authorities to make decisions on the number of meals to prepare in a day. Besides, it will be possible to predict demand by considering the submitted request and history of the cafeteria. For example, using random forest regression and XGBoost Regression algorithms, one can predict demand based on the gathered data and current requirements. The results can then be compared with a threshold to get an approximate target quantity to prepare. The proposed approach will be innovative by considering the student data to come up with a new way of estimating the meal demand in a cafeteria. As a result, the administration will address the problem of excess preparation over the expected quantity of students. Therefore, it will help reduce wastage and use of electricity in a school. Another strength of the proposed solution is that it will help manage the surplus. At the time of preparation, the system will support the management in recording the excess meals as the number of servings to students. Thus, the suggested solution will solve the prediction and management of surplus in a cafeteria. A food-waste management system in a cafeteria benefits tremendously from a prediction model. There is a need for administrative staff to develop an appreciation of the power of historical data in a cafeteria. First, it is important to record all meals prepared and served to help create a statistical model for the new system. Secondly, the administrator should consider recording the data on holidays, examinations, and class attendance schedules. Besides, weather conditions are also a good predictor of the demand for meals in a cafeteria.

In the light of future developments, FoodPulseAI can be developed further by using a wider range of historical demand data sets to increase accuracy of the predicted demand. More advanced machine-learning and deep learning techniques can also be explored and compared with the proposed algorithm other than the proposed Random Forest and XGBoost. Dynamic factors such as weather, attendance rates, examinations, holidays, and campus events can be included in the prediction system.

The system can also be expanded by integrating a mobile application to support student voting and administrative supervision. Automated student reminders can also be included which can inform students of voting periods and alert cafeteria staff of significant demand changes. Similarly, the system can be programmed to automatically send notifications to registered NGOs in the event of an occurrence of surplus food that can be redistributed.

Lastly, FoodPulseAI can implement a self-actualizing system that updates the history demand database with consumption and surplus figures after every meal cycle, essentially

teaching the system from its previous errors. In addition, the system can be expanded at the department, hostel, cafeteria, and institutional level to monitor and compare food wastage and demand figures generating valuable insights to enhance food management practices.

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